ASPIRE • Amplicon analysis with Nextflow

Reads → filtered ASVs → prepared tables → analyses → reports

Inputs

FASTQs • Manifest • Metadata • References
Optional: controls • VOCs • Genomes

Module · Compute · Data · Input · Output

FASTQ → ASVs

Paired FASTQs → Primer removal / read QC (Cutadapt / fastp) → Merge / filter (VSEARCH) → Denoise / remove chimeras / count (VSEARCH) → Initial ASVs + counts

ASV filtering

Initial ASVs + counts → Taxonomy (SINA / QIIME 2) → Depth QC / control decontamination (decontam (TECH / BIO)) → Non-target removal / abundance / prevalence (MITOMASTER / BLAST+) → Filtered ASVs + counts

Metadata & transformations

ASVs + metadata → Link sample metadata → Transform counts / batch correction (ConQuR UMAP / HDBSCAN) → Group diagnostics / cohort selection (scikit-learn) → Counts / CLR + metadata

Community analysis

Prepared tables

- Full biological cohort → Diversity / ordination (scikit-bio / vegan)
- Selected cohort → Indicators / ASV clustermaps contrasts / power (indicspecies / vegan)
- Selected cohort → Networks / genome links (SPIEC-EASI / BLAST+ igraph / NetworkX)

→ Community results

ASV–VOC integration

ASVs / VOCs / ISA + sample metadata → Match samples and measurements → Exploratory sample-level Spearman correlations / Patient-level inference permutations / FDR → ISA overlays / VOC–ASV clustermaps → Association tables and heatmaps

Summaries & reports

Results + audits → Read statistics / result summaries (SeqKit) → Sankey / report → Logs / checksums (Nextflow) → Tables / figures + provenance

Optional steps follow configuration. CLR: centered log-ratio • ISA: indicator species analysis.